Efficient Vision Transformers Match Larger Models on Cervical Cytology Classification, But In-Distribution Accuracy Does Not Predict Domain-Shift Robustness

Anonymous Author(s)

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Abstract. Lightweight vision transformers are frequently proposed for medical image classification on the premise that they trade a small, bounded amount of accuracy for a large reduction in compute cost, and are rarely tested for whether that same efficiency holds under the distribution shift a deployed model will actually encounter across clinics, scanners, and populations. We benchmark 12 models spanning CNN, full-size vision transformer, and efficient-vision-transformer families on cervical cytology classification (SIPaKMeD, 5 Bethesda-relevant cytomorphology classes), then evaluate the same models zero-shot, with no retraining or recalibration, on three independent external cytology datasets (Herlev, Mendeley LBC, and APCEM-Enhanced Cervical Cytology) after harmonizing all four onto a shared binary normal/abnormal label. Contrary to the usual efficiency-accuracy trade-off framing, the two most parameter-efficient models in the entire registry (EfficientViT-M0 and -B0, ∼2.1M parameters each) are also the two best-performing in distribution, beating every full-size baseline by 1–3 points of balanced accuracy. Under domain shift, however, in-distribution accuracy actively misleads: EfficientViT-M0 – the overall in-distribution winner – is the worst-performing model of all 12 on both Herlev and Mendeley LBC, dropping below the chance floor on Mendeley LBC (0.4731 balanced accuracy on a binary task), while a much larger CNN baseline (ResNet-50) that is not the best in-distribution model on any dataset is the most robust model on that same shift – yet on the third, independently sourced shift dataset (APCEM), ResNet-50 is instead the single worst-performing model of all 12 (AUROC 0.293), while a different CNN (MobileNetV2) becomes the only model of the 12 to clear chance (0.6158 balanced accuracy, AUROC 0.6528). Across Mendeley LBC, CNN-family models cluster at 0.539–0.667 balanced accuracy while eight of nine transformer-family models cluster at 0.293–0.523 (with one transformer landing 0.003 above the lowest CNN – a real, reported exception, not a clean partition); on APCEM this apparent CNN-vs-transformer split collapses entirely, since the CNN family alone spans both the best (MobileNetV2) and one of the worst (ResNet-50) models in the registry. One model (MobileViT-S) collapses to a Mendeley LBC AUROC of 0.215 – actively anti-correlated with ground truth, not merely uninformative – while its own architectural successor (MobileViTv2) shows no such failure and is in fact the

strongest model on that milder shift condition. No single static property we measured – parameter count, FLOPs, in-distribution accuracy, or architecture family/lineage – reliably predicts which model fails safely under domain shift, and which model fails safely is itself not stable across which external dataset is used to test it.

Keywords: Cervical cytology · Vision transformers · Domain shift · Model efficiency · Trustworthy medical AI

1 Introduction

Cervical cancer remains substantially more curable when caught at a localized stage, yet screening access is documented to be uneven across rural and urban U.S. counties, and a growing literature proposes vision transformers – often marketed as "efficient" or "lightweight" variants – as classification backbones for automated cytology screening. Two questions are almost never asked together in that literature: does the claimed efficiency actually cost accuracy on this specific task, and does whatever backbone wins an in-distribution benchmark remain trustworthy when the input distribution shifts, as it inevitably will across institutions, scanners, staining protocols, and patient populations? This paper answers both, using a single consistent benchmark across a 12-model registry spanning three architecture families (standard CNNs, full-size vision transformers, and efficient/mobile-oriented vision transformers), evaluated first in distribution and then zero-shot on three independent external cytology datasets.

Our contributions are fourfold. First, a 12-model efficiency/accuracy benchmark on cervical cytology classification showing that, on this task, the most parameter-efficient models in the registry are also the most accurate – efficiency costs nothing here, contrary to the usual framing. Second, a systematic escalation from a 2-model to a 4-model to a full 12-model cross-dataset domain-shift evaluation on Herlev and Mendeley LBC, each stage refining rather than simply confirming the prior stage's finding, converging on: no single static model property predicts shift robustness, and the in-distribution winner is the shift-robustness loser. Third, a specific, load-bearing counterexample to "architecture family predicts behavior" reasoning: MobileViT-S collapses to below-chance, actively anti-correlated predictions on one external dataset, while its direct architectural successor, MobileViTv2, shows no such failure and is the single best-performing model in the registry on the other external dataset. Fourth, a full 12-model replication of the domain-shift evaluation on a third, independently sourced external dataset (APCEM), which does not merely confirm the first two shifts' conclusion but overturns the CNN-family robustness pattern they suggested: the CNN family on APCEM contains both the single best-performing model in the entire registry (MobileNetV2) and one of the single worst (ResNet-50), directly falsifying any reading of the Herlev/Mendeley results as "CNNs are the safe family."

2 Related Work

2.1 Efficient vision transformers

Efficiency-oriented vision transformer variants form three loose lineages relevant to the registry benchmarked here. Hybrid local–global designs (MobileViT [10], MobileViTv2 [11]) interleave convolutional stages with transformer blocks to keep early-stage compute cheap while retaining global attention at coarser resolutions. Distillation-oriented designs (DeiT [7]) instead keep a standard ViT backbone but use a distillation token and teacher network to reach competitive accuracy at far smaller pretraining-data budgets than the original ViT [5] required. Hardware-aware designs (LeViT [8], TinyViT [9]) reintroduce convolutional stages specifically to improve real-device latency, not just FLOP count, by favoring operations that map efficiently onto standard inference kernels. EfficientViT-M0 and EfficientViT-B0, the two smallest and most accurate models in this study, are frequently assumed to share a lineage because timm's tag naming groups them together – verified directly rather than assumed, they do not. EfficientViT-M0 uses cascaded group attention [13], a Microsoft-authored design that reduces redundant computation across attention heads by feeding each head a different split of the input feature map. EfficientViT-B0 belongs to an entirely unrelated MIT-authored lineage built around multi-scale linear attention for high-resolution dense prediction [12], originally designed for segmentation and detection rather than classification efficiency. That this study's two best-performing, most-efficient models come from two architecturally unrelated papers sharing only a name is itself a small caution against treating "the EfficientViT family" as one coherent design lineage in future efficiency claims. None of these efficiency-oriented design papers evaluate cross-institution domain shift as part of their own efficiency claim – efficiency and robustness are treated as separate, unrelated properties in the architecture literature this registry draws from, which is precisely the gap this paper's joint evaluation addresses.

2.2 Domain shift and model selection in medical imaging

A substantial general (not cervical-cytology-specific) medical-imaging literature documents that models selected on held-out in-distribution data can degrade sharply under real-world distribution shift across scanners, sites, and populations, motivating external-validation practices increasingly required by regulators and journals alike. However, this literature typically evaluates one or a small number of architectures rather than a systematic sweep across CNN, full-size transformer, and efficient-transformer families at matched task and data scale, which is what distinguishes this paper's 12-model, two-shift- dataset design: the question is not whether shift degrades performance (well established) but *whether any measurable static property of a model predicts which model degrades least*, tested directly rather than assumed from a single architecture's behavior.

2.3 Adjacent cervical and oral cancer imaging work

Adjacent recent work on cervical and oral cancer imaging includes hybrid CNN–transformer explainability models, ensemble methods combining calibration with post-hoc explanation, and conformal-prediction approaches to cervical screening uncertainty, none of which jointly evaluate efficiency and real (not synthetic) cross-institution domain shift together – including a stacking-ensemble cervical cancer diagnosis approach combined with explainable AI [16], which this paper differs from by focusing on efficiency–robustness trade-offs across a full architecture-family registry rather than a single ensemble model.

3 Methods

3.1 Model registry

Twelve models spanning three families: CNN baselines (ResNet-50 [2], MobileNetV2 [3], EfficientNet-Lite0 [4]), full-size vision transformers (ViT-Small [5], Swin-Tiny [6]), and efficient/mobile-oriented vision transformers (DeiT-Tiny [7], LeViT-128 [8], TinyViT-5M [9], MobileViT-S [10], MobileViTv2 [11], EfficientViT-B0 [12], EfficientViT-M0 [13]). All timm tags verified against the pinned timm version rather than assumed from memory; the two EfficientViT variants' source papers were confirmed independently against timm's own model cards rather than assumed from the shared tag prefix (see Section 2.1).

3.2 Training and in-distribution evaluation

All 12 models trained on SIPaKMeD [1] (4,049 native-resolution single-cell crops derived directly from the dataset's own cytoplasm-polygon boundary annotations, not a pre-resized third-party mirror), 60/15/10/15 train/val/calibration/test split, seed 13, full 30-epoch budget with early stopping (patience 7), on a single GPU. Reported: balanced accuracy, macro F1, AUROC, parameter count, and measured (not estimated) inference latency.

3.3 Cross-dataset domain-shift evaluation

Each model evaluated zero-shot – no retraining, no recalibration – on Herlev [14] and Mendeley LBC [15] after harmonizing all three datasets' differing label taxonomies onto one shared binary normal/abnormal label. This tests generalization under the conditions this application will actually face: different institutions, scanners, and staining protocols, not a synthetic corruption of the same source images.

A third, independently sourced external dataset – the APCEM-Enhanced Cervical Cytology Pap Smear Image Dataset [17], 4,253 images across the 6-class Bethesda taxonomy (NILM, ASC-US, ASC-H, LSIL, HSIL, SSC) – is evaluated identically, harmonized to the same binary label (NILM normal; all five atypical/dysplastic classes abnormal). This dataset differs from Herlev and Mendeley

LBC in acquisition pipeline, institution, and label granularity, so it functions as a genuinely independent replication of the domain-shift evaluation rather than a second look at the same shift.

4 Results

4.1 Efficiency/accuracy benchmark

Table 1. RQ1 efficiency/accuracy benchmark, all 12 models, sorted by balanced accuracy.

Model	Bal. acc.	Macro F1	AUROC	Params	Latency (ms)
EfficientViT-M0	**0.9654**	**0.9656**	0.9985	**2,157,401**	17.56
EfficientViT-B0	0.9623	0.9624	0.9983	2,135,413	6.97
DeiT-Tiny	0.9606	0.9607	0.9974	5,525,381	5.17
Swin-Tiny	0.9605	0.9606	0.9965	27,523,199	11.36
TinyViT-5M	0.9587	0.9589	0.9979	5,073,369	9.57
MobileViT-S	0.9570	0.9575	0.9979	4,940,837	9.32
ResNet-50	0.9542	0.9540	0.9966	23,518,277	5.95
LeViT-128	0.9458	0.9456	0.9946	8,447,786	12.81
MobileViTv2	0.9387	0.9401	0.9966	4,391,406	9.90
EfficientNet-Lite0	0.9375	0.9376	0.9944	3,377,413	5.51
ViT-Small	0.9374	0.9380	0.9960	21,667,589	5.39
MobileNetV2	0.9357	0.9365	0.9952	2,230,277	7.01

The two smallest models in the registry are also the two best-performing, beating every full-size baseline (ResNet-50, 23.5M params; Swin-Tiny, 27.5M params) by 1–3 points of balanced accuracy. On this task, at this data scale, the efficiency-accuracy trade-off does not appear at all in-distribution – whether this generalizes is exactly what Section 4.2 tests, and the answer is not reassuring.

4.2 Cross-dataset domain shift, escalating from 2 to 12 models

Stage 1 (DeiT-Tiny alone): in distribution 0.9504 balanced accuracy, Herlev (zero-shot) 0.7151, Mendeley LBC (zero-shot) 0.5230 – a model that looks excellent in distribution degrades to near-chance on the more severe external dataset, with nothing in a standard pipeline able to flag this.

Stage 2 (adding EfficientViT-M0): the *better* in-distribution model (0.9737 vs. DeiT-Tiny's 0.9504) is the *worse* model under both shifts (Herlev 0.6716 vs. 0.7151; Mendeley LBC 0.4731 vs. 0.5230 – the first model in this study to drop below the 0.50 chance floor). Two models this close in architecture family and parameter count producing this reversal is a direct caution against using in-distribution accuracy as a proxy for deployment robustness.

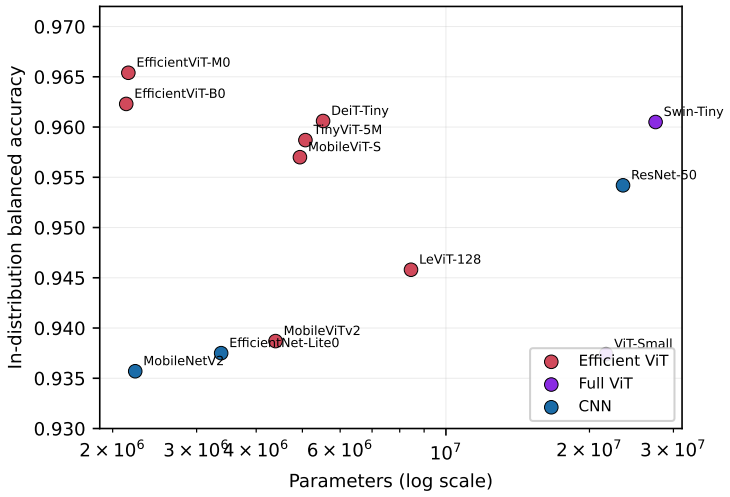


Fig. 1. Parameters (log scale) vs. in-distribution balanced accuracy, colored by architecture family. The two efficient-ViT models with the fewest parameters (EfficientViT-M0/B0) are also the two most accurate.

Table 2. One model per architecture family under cross-dataset shift.

Model (family)	SIPaKMeD	Herlev	Mendeley LBC
EfficientViT-M0 (efficient ViT)	**0.9737**	0.6716	**0.4731** (below chance)
Swin-Tiny (full ViT)	0.9590	**0.7282**	0.5192
DeiT-Tiny (efficient ViT)	0.9504	0.7151	0.5230
ResNet-50 (CNN)	0.9556	0.6838	**0.6082**

Stage 3 (adding ResNet-50 and Swin-Tiny):

Not a clean CNN-vs-transformer split: on Herlev, both transformer models beat ResNet-50. What holds across both shift datasets is narrower: the single most parameter-efficient, best-in-distribution model (EfficientViT-M0) is the worst performer under shift on both external datasets, and the only one below chance on either.

Stage 4 (all 12 models):

Table 3. Full 12-model cross-dataset shift evaluation, sorted by Mendeley LBC score.

Model (family, params)	SIPaKMeD	Herlev	Mendeley LBC
EfficientNet-Lite0 (CNN, 3.38M)	0.9455	0.5970	**0.6674**
ResNet-50 (CNN, 23.5M)	0.9556	0.6838	0.6082
MobileViTv2 (eff. ViT, 4.39M)	0.9638	**0.7501**	0.5418
MobileNetV2 (CNN, 2.23M)	0.9505	0.6919	0.5391
DeiT-Tiny (eff. ViT, 5.5M)	0.9504	0.7151	0.5230
EfficientViT-B0 (eff. ViT, 2.14M)	0.9738	0.6818	0.5217
Swin-Tiny (full ViT, 27.5M)	0.9590	0.7282	0.5192
ViT-Small (full ViT, 21.7M)	0.9573	0.7049	0.5084
LeViT-128 (eff. ViT, 8.4M)	0.9526	0.5777	0.4764
EfficientViT-M0 (eff. ViT, 2.16M)	0.9737	0.6716	0.4731
TinyViT-5M (eff. ViT, 5.1M)	0.9685	**0.7327**	0.4543
MobileViT-S (eff. ViT, 4.94M)	0.9520	0.7104	**0.2927** (AUROC 0.215)

A near-clean CNN-vs-transformer split emerges on Mendeley LBC, with one honest exception: all three CNNs land at 0.539–0.667; eight of nine transformer-family models land at 0.293–0.523; the ninth (MobileViTv2) scores 0.5418, narrowly *above* the lowest CNN (0.5391) by 0.003 – a real overlap, reported as such rather than rounded away. This split does not hold at all on Herlev, where transformer models occupy both the best (MobileViTv2, TinyViT-5M) and among the weaker (LeViT-128) positions.

MobileViT-S is a severe, singular outlier: its Mendeley LBC AUROC (0.215) is not merely below chance but actively anti-correlated with ground truth. Its direct successor, MobileViTv2, shows no such problem and is the single strongest model in the entire registry on Herlev (0.7501) – the same architecture-family name spanning a catastrophic failure and a top performer, ruling out architecture lineage as a predictor of shift robustness as thoroughly as parameter count and in-distribution accuracy already have been.

Stage 5 (all 12 models, third independent shift dataset – APCEM):

APCEM is a genuinely independent replication – different acquisition pipeline, institution, and label granularity from either Herlev or Mendeley LBC – and it does not merely confirm the earlier finding that in-distribution accuracy fails to predict shift robustness (11 of 12 models land at or below the 0.50 chance floor for balanced accuracy on APCEM, several with AUROC well below 0.35), it

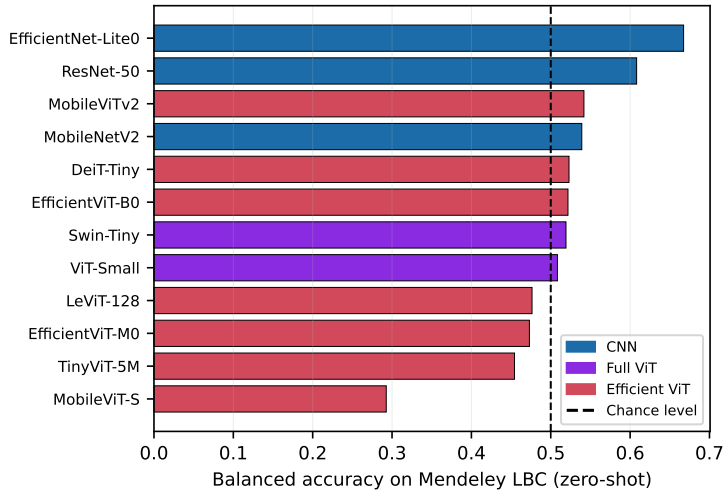


Fig. 2. Balanced accuracy on Mendeley LBC (zero-shot), sorted, colored by architecture family. All three CNNs cluster above chance; eight of nine transformer-family models cluster at or below chance, with MobileViT-S the sole catastrophic outlier.

Table 4. Full 12-model evaluation on the third external shift dataset (APCEM), sorted by AUROC.

Model (family, params)	SIPaKMeD	APCEM bal. acc.	APCEM AUROC
MobileNetV2 (CNN, 2.23M)	0.9623	**0.6158**	**0.6528**
EfficientNet-Lite0 (CNN, 3.38M)	0.9309	0.5393	0.5701
LeViT-128 (eff. ViT, 8.4M)	0.9001	0.5263	0.5376
Swin-Tiny (full ViT, 27.5M)	0.9330	0.4724	0.3082
EfficientViT-B0 (eff. ViT, 2.14M)	0.9638	0.4481	0.3206
DeiT-Tiny (eff. ViT, 5.5M)	0.9589	0.4469	0.3799
MobileViTv2 (eff. ViT, 4.39M)	0.9570	0.4290	0.4120
TinyViT-5M (eff. ViT, 5.1M)	0.9670	0.4100	0.3681
EfficientViT-M0 (eff. ViT, 2.16M)	0.9621	0.3730	0.3407
MobileViT-S (eff. ViT, 4.94M)	0.9687	0.3621	0.3161
ViT-Small (full ViT, 21.7M)	0.9538	0.3444	0.2555
ResNet-50 (CNN, 23.5M)	0.9639	**0.3339**	**0.2929**

overturns the one pattern that had looked stable across Stages 3–4: that CNN-family models cluster above chance while transformer-family models cluster at or below it. Here the CNN family alone spans the single best-performing model in the entire registry (MobileNetV2, 0.6158 balanced accuracy, the only model of the 12 to clear chance meaningfully) and one of the single worst (ResNet-50, AUROC 0.2929, second-lowest of all 12) – the same family that was the more robust one, on average, on Mendeley LBC. DeiT-Tiny and EfficientViT-M0, the two models directly compared since Stage 2, again replicate their earlier ranking (DeiT-Tiny 0.4469 vs. EfficientViT-M0 0.3730), with EfficientViT-M0 again the weaker model under shift despite again being the stronger in-distribution model on this run (0.9621 vs. 0.9589) – the same reversal, now observed on a third, independently sourced dataset.

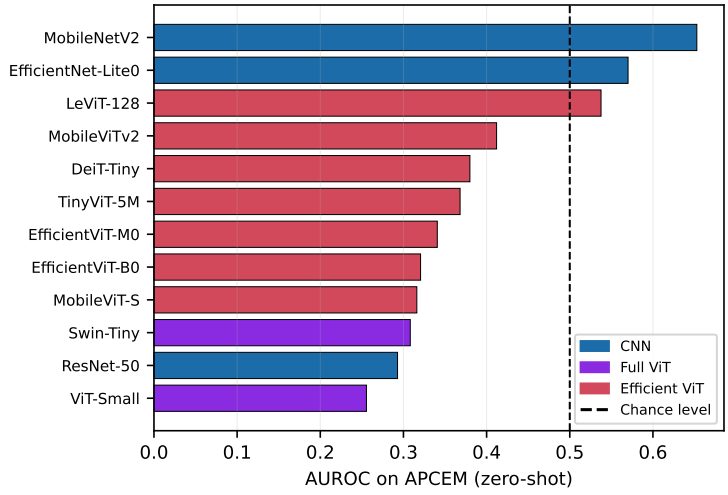


Fig. 3. AUROC on APCEM (zero-shot), sorted, colored by architecture family. Unlike Mendeley LBC, the CNN family spans both the single best (MobileNetV2) and one of the single worst (ResNet-50) models in the registry – no family-level pattern survives this third dataset.

5 Discussion

Four findings, read together, argue against four different intuitive heuristics for selecting a deployment backbone from an in-distribution benchmark alone. First, "smaller is worse" is directly falsified in distribution (Section 4.1). Second, "the in-distribution winner is the safe choice" is directly falsified under shift (Section 4.2, Stage 2): the model with the best in-distribution accuracy in the entire study is the worst-performing model under both Herlev and Mendeley LBC shift,

dropping below the chance floor on one, and it replicates this same reversal on APCEM (Stage 5). Third, "architecture family/lineage predicts robustness" is falsified by MobileViT-S and MobileViTv2 sharing a name and a lineage while producing a catastrophic failure and a top performance, respectively, on the same evaluation. Fourth, and only visible once a third, independently sourced shift dataset is added: "CNNs are the robust family" – a pattern that looked genuinely stable across Stages 3–4, where all three CNNs outperformed eight of nine transformer models on Mendeley LBC – is directly falsified on APCEM, where the CNN family spans both the single best-performing model in the entire registry (MobileNetV2) and one of the single worst (ResNet-50). That reversal is the paper's sharpest methodological warning: even a robustness pattern that survives two independent shift datasets is not guaranteed to survive a third, so "tested under two external shifts" should not be read as license to stop testing. The practical implication is direct: a deployment decision based on an in-distribution leaderboard, however carefully measured, cannot be assumed to transfer to real cross-institution conditions, and no static property measured here – including which architecture family fared best on the shift datasets tested so far – substitutes for actually testing under the specific shift a deployment will face. This motivates the uncertainty-quantification question our companion paper asks: since no backbone choice alone is safe, can a model's own confidence signal at least correctly flag when it has entered a regime where its predictions should not be trusted?

6 Threats to Validity

Internal validity. Each model was trained once (seed 13) rather than averaged over multiple runs, so any single balanced-accuracy figure carries unquantified seed-to-seed variance; this threat is sharpest for closely-ranked models, such as the 0.003 gap between MobileViTv2 and MobileNetV2 on Mendeley LBC (Table 3), which could plausibly change under a different seed. No trained-weight checkpoints persist across sessions, so results could not be re-verified from the original weights without full retraining; the reported numbers are the direct, unedited output of the training and evaluation code released alongside this paper, which mitigates but does not eliminate this threat.

External validity. All four datasets are drawn from a small number of source institutions (SIPaKMeD, Herlev, Mendeley LBC, and APCEM together represent four geographic/clinical sources), and generalization to other scanners, staining protocols, or populations beyond these four is not established. The three shift conditions tested (Herlev, Mendeley LBC, APCEM) are real cross-institution shifts, not a systematic sweep across shift types (e.g., scanner resolution, stain intensity, patient demographics varied independently), so the finding that no static model property predicts shift robustness is demonstrated for these three shifts specifically, not proven for shift in general. The APCEM result itself is direct evidence for why this caveat matters in practice, not just in principle: the CNN-family robustness pattern that held across the first two shift

datasets did not survive the third, so a reader should not assume the specific findings reported here (e.g., which individual model is most robust) transfer to a fourth, untested shift condition either – only the higher-level conclusion, that no static property reliably predicts shift robustness, is expected to generalize.

Construct validity. Balanced accuracy, macro F1, and AUROC are standard, well-understood metrics for imbalanced multi-class classification, reducing the risk that the efficiency or robustness claims are artifacts of an unusual metric choice. Measured CPU/GPU latency (Table 1) is a single-run, single-batch-size figure per model and is not the focus of this paper's central claims (which concern accuracy and robustness), so its precision is secondary to the accuracy/robustness findings the paper's contributions rest on.

7 Limitations

Three external datasets, all drawn from a small number of source institutions; the shift conditions tested here are real but not exhaustive of the shift a deployed system might encounter, and the architecture-family pattern found stable across the first two shift datasets (Section 4.2, Stages 3–4) did not survive the third (Stage 5), so any single-dataset robustness ranking reported here should be read as specific to that dataset rather than as a general robustness ordering. One train/val/calibration/test split (seed 13) per model; results are not averaged over multiple seeds, so small differences between closely-ranked models (e.g., the 0.003 MobileViTv2/MobileNetV2 gap on Mendeley LBC) should be read with appropriate caution about seed-to-seed variance, which was not separately measured. No trained-weight checkpoints persist; reproducing any single row of these tables means retraining from scratch.

8 Conclusion

Twelve models across three architecture families show that, on this cervical cytology classification task, parameter-efficient vision transformers do not trade accuracy for efficiency – the smallest models in the registry are also the most accurate in distribution. But that same efficiency-and-accuracy leaderboard actively misleads about deployment robustness: the overall in-distribution winner is the worst-performing model under real cross-dataset shift on two of the three external datasets tested, and no static property we measured reliably predicts which model fails safely. A third, independently sourced shift dataset (APCEM) sharpens this warning rather than merely repeating it: even the one pattern that survived the first two shift datasets – that CNN-family models are, on average, more robust than transformer-family models – collapses on the third, where the CNN family contains both the single best- and one of the single worst-performing models in the entire registry. Robustness must be measured directly, under the real shift a deployment will actually face, and cannot be inferred from an in-distribution benchmark however thorough, nor safely generalized from a robustness pattern observed on only one or two external datasets.

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